

Dr. Thejus Mahajan
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Hamburg, 10 March 2026

Prof. Dr. Matthias Schwab & Dr. Elke Schaeffeler
Dr. Margarete Fischer-Bosch Institute of Clinical Pharmacology
Robert-Bosch-Krankenhaus
Auerbachstraße 112
70376 Stuttgart

Post Doctoral Research Scientist - Bioinformatics (m/f/d)

Dear Prof. Dr. Schwab and Dr. Schaeffeler,

My experience refactoring production-grade medical data pipelines has taught me how crucial robust computational infrastructure is for clinical research. The opportunity to elucidate the tumor microenvironment by integrating spatial transcriptomics and scRNA-seq data at your institute directly matches this expertise. My transition from a Ph.D. in computational systems modeling to bioinformatics pipeline engineering makes me a highly suitable candidate to develop the novel computational solutions your clinical research demands.

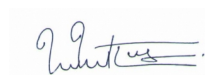
While my academic background is in systems modeling, I have spent the last year intensively retraining in modern bioinformatics, transcriptomics, and machine learning. At my current internship with HealthTwist GmbH, I am solely responsible for refactoring a production medical data pipeline (in R/tidyverse) for Germany's interventional radiology quality registry. Working with 143,000 patient records and 1,920 variables, I externalized hardcoded logic into configuration files, verified byte-level reproducibility, and reduced the 1,834-line codebase by 26%—establishing the robust biostatistics foundation required for integrating clinical phenotypes with multi-omics data.

Your project's need to bridge single-cell datasets with clinical outcomes requires deep expertise in both programming and data integration, which are my core strengths:

- **Bioinformatics & Transcriptomics Focus:** Highly proficient in building reproducible, containerized workflows (Nextflow/DSL2) and processing multi-omic data. I bring a strong understanding of the statistical modeling required to extract signal from high-dimensional biological datasets like spatial transcriptomics and scRNA-seq.
- **Strong Programming in R and Python:** I possess advanced programming skills across both ecosystems. In R, I build clinical data workflows (Tidymodels, Bioconductor, tidyverse) and perform complex biostatistics. In Python, I have engineered data analysis pipelines using Pandas, scikit-learn, and high-performance array frameworks (JAX), giving me the tools needed to prototype new computational solutions for data integration.
- **Large-Scale Data Integration & Oncology Interest:** Processed terabytes of raw experimental data during my Ph.D. and developed complex algorithms for HPC clusters. I am eager to apply my expertise in navigating coupled systems to cancer biology, specifically understanding metastasis and the tumor microenvironment in primary renal cell carcinoma.

I am deeply motivated by the Institute of Clinical Pharmacology's mission to bridge omics data with clinical phenotypes. My robust mathematical background and modern data engineering skills will allow me to immediately contribute to your interactive team of bioinformaticians, clinicians, and biologists. I welcome the opportunity to discuss further and am available for an interview.

Sincerely



Dr. Thejus Mahajan

THEJUS MAHAJAN PH.D.

Bioinformatics & Data Scientist

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Hamburg, Germany

*30.09.1992 (Talikulam, India)

Married



PROFILE

- **Focus:** Transcriptomics data analysis, multi-omics integration, and bioinformatics pipeline development.
- **Core Stack:** R (Bioconductor, Tidymodels), Python (Pandas, JAX, Scikit-learn), Nextflow (DSL2), BASH.
- **Current Objective:** Applying rigorous mathematical and programming skills in single-cell and spatial transcriptomics to advance research in oncology and tumor biology.

EXPERIENCE

Internship – Bioinformatics & Data Pipeline Engineering

HealthTwist GmbH

02/2026 - 04/2026

Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn

- Refactored a 1,834-line medical data pipeline (R/tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143K rows × 1,920 columns.
- Externalized 257 hardcoded rules into 8 config files; 26.5% code reduction with byte-level output verification (`identical()`).
- Identified 2 latent bugs through systematic code analysis; documented without altering pipeline behavior.
- Machine Learning workflows using the Tidymodels framework in R.

Further Training – Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH

08/2025 - 02/2026

Berlin, Germany

Application-related Bioinformatics and Biostatistics

- Programming methodology: Shell (BASH), Python (object-oriented), R.
- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Nextflow, Galaxy); structural bioinformatics.
- Clinical biostatistics (R/Bioconductor, ANOVA, PCA, HCA, survival analysis); GDPR/BDSG compliance.

Guest Scientist

Helmholtz-Zentrum Hereon, Ökosystemmodellierung

05/2025 - 10/2025

Geesthacht, Germany

Computational Biologist & Systems Modeler

- Spearheaded research modeling evolutionary processes in marine ecosystems.
- Applied advanced computational techniques to simulate long-term environmental impacts.

Job Search & German Language Learning

Hamburg, Germany

02/2025 - 04/2025

Hamburg, Germany

Active job search and German language acquisition (Goethe B1 certification).

Post-doctoral Scientist

University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences

08/2021 - 01/2025

Hamburg, Germany

Marine Ecosystem Modelling

STRENGTHS



Data Pipeline Engineering

Refactored a production medical data pipeline (1,834 lines, 143K rows), achieving 26.5% code reduction with zero behavioral regression. Byte-level verification after every change.



Complex Problem-Solving

Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector, recalculating results using multi-sensor data.



Bioinformatics & Biostatistics

Bridged physical science expertise with modern biological workflows. Skilled in complex multidimensional array operations, JAX/GPU acceleration, sequence alignment, and multivariate statistical modeling for multi-omics using R (Bioconductor) and Python.

TECHNICAL SKILLS

Programming

Python (JAX, Pandas, NumPy, Scikit-learn, Biopython), R (Bioconductor, Tidymodels), SQL, BASH

Bioinformatics

Nextflow (DSL2), scRNA-seq & Spatial Transcriptomics Concepts, NGS Analysis, Multiple Sequence Alignment (MAFFT), Sequence Trimming (trimAI), NCBI / Entrez

Biostatistics & ML

Deep Learning Foundations (TensorFlow, PyTorch concepts via JAX), Multivariate Analysis, PCA, Clustering, Scikit-learn, Tidymodels

Data Engineering

Hardware Acceleration (TPU/GPU), Large dataset processing (HDF5, NetCDF), Pipeline development

Tools

Linux/Unix, Git, HPC clusters, Conda

TRAINING / COURSES

IBM: Machine Learning with Python; Introduction to AI

Coursera (in progress)

High Performance Computing (HPC) Training

Jülich Supercomputing Center (JSC), Forschungszentrum Jülich

- Model Development: Developed the novel “Cyanobacteria Life Cycle” (CLC) model within the ERGOM framework for climate warming projections.
- High-Performance Computing & ML: Led the translation of a legacy Fortran continuous model into a high-performance Python simulation engine. Utilized **Google JAX** vectorization and branchless logic for massive parallelization (direct compatibility with **TPUs and GPUs**).
- This architecture established a strong foundation in tensor operations underpinning deep learning frameworks like **PyTorch** and **TensorFlow**.

Job Search

Kerala, India

04/2021 - 07/2021

Kerala, India

Relocation from India to Germany and active job search.

Physics Educator

Tutorwaves Solutions Inc / Freelance

10/2018 - 03/2021

Kerala, India

Prepared physics content for competitive examinations and coached state-level chess teams.

Ph.D. Researcher in Astrochemistry

University of Paris-Saclay

10/2015 - 09/2018

Orsay, France

Led atomic collision experiments, analyzed data, and modeled outcomes.

- Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
- C++ signal processing, algorithm development, and molecular fragmentation modeling (KIDA database).

EDUCATION

Ph.D. in Astrochemistry

Université Paris-Saclay, Institute of Molecular Sciences (ISMO) Orsay, France

09/2015 - 09/2018

Orsay, France

M.Sc. in Physics

National Institute of Technology Calicut, India

07/2012 - 01/2015

Calicut, India

B.Sc. in Physics

University of Calicut, India

06/2009 - 04/2012

Thrissur, India

REFERENCES

Dr. Andreas Busjahn

abusjahn@healthtwist.de

Prof. Dr. Kai Wirtz

kai.wirtz@hereon.de

Prof. Dr. Elisa Schaum

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Dr. Thejus Mahajan

MITx: 7.00x Introduction to Biology - The Secret of Life

Comprehensive coursework on genetics, biochemistry, and molecular biology.

OTHER SELECTED PROJECTS

Hepatitis Delta Virus Pipeline • GitHub

Nextflow (DSL2) pipeline for viral sequence analysis: NCBI Entrez download, MAFFT alignment, trimAl trimming, Conda-managed dependencies.

JAX/TPU High-Performance Simulator • GitHub

Ported a legacy Fortran model to a modular Python architecture using Google JAX. Engineered branchless, vectorized logic for massive TPU/GPU parallelization.

Introduction to Tidymodels • GitHub

Teaching script for ML in R: recipes, parsnip, workflows, and yardstick on the Palmer Penguins dataset.

Berlin Tree Analysis • GitHub

Interactive Jupyter notebook analyzing 5 years of tree planting data in Berlin using Python, Pandas, and Seaborn.

PUBLICATIONS

5 peer-reviewed articles • Google Scholar Profile

LANGUAGES

English



Proficient

German



B1 - Goethe Certified

French



Intermediate

Malayalam



Native

Tamil



Proficient

Hindi



Intermediate